

Level 2: Intermediate – Submission Report

Project: TaskFlow – Full Stack Task Management System

Organization:	Cognifyz Technologies	Domain:	Full Stack Web Development
Status:	Completed & 100% Verified	Author / Intern:	Full Stack Intern (vikast2849)
Tasks Included:	Task 3 & Task 4	Repository:	github.com/vikast2849/taskflow

1. Level Overview & Objectives

- **Task 3 (Advanced CSS Styling & Responsive Design):** Professional 5-section layout (Navbar, Hero, Stats, Task Cards, Footer), CSS custom properties, glassmorphism, hover animations, and Bootstrap 5 responsive grid.
- **Task 4 (Complex Validation & Dynamic DOM SPA):** Modular ES architecture (validation, taskManager, router, app), live 5-tier password complexity analyzer with progress meter, and client-side hash routing.

2. Route Endpoints & Verification URLs

- **Task 3 Responsive Layout:** /tasks/task-3 – Multi-section dashboard demonstrating CSS variables, glassmorphism, and responsive breakpoints.
- **Task 4 Dynamic DOM SPA:** /tasks/task-4 – Single Page Application with dynamic DOM task CRUD, real-time search/filter, and password strength meter.

3. Implementation Files & Technical Architecture

Component / File Path	Key Technical Functionality
server/controllers/task3Controller.js	Prepares sprint velocity metrics and multi-card showcase data.
public/css/task3.css	Advanced CSS variables, glassmorphism blur, hover lifts, and keyframe animations.
views/tasks/task-3.ejs	Responsive 5-section template tested across mobile, tablet, and desktop.
public/js/task4/validation.js	RFC 5322 email regex and 5-criterion live password strength analyzer.
public/js/task4/taskManager.js	In-memory dynamic DOM task operations (add, status toggle, animated delete, search, filter).
public/js/task4/router.js	Hash-based client router (#tasks, #create, #register-demo, #stats).
public/js/task4/app.js	Orchestrator synchronizing DOM mutations with live statistics recalculation.
tests/test_task3.js & test_task4.js	Automated test scripts verifying responsive assets and modular SPA scripts.

4. Submission Summary for Internship Portal

Cognifyz Level 2 (Intermediate) Submission:

- Completed Tasks 3 & 4 in full compliance with Cognifyz internship guidelines.
- Designed responsive 5-section layout using CSS Grid, Flexbox, glassmorphism, and animations.
- Engineered clean ES module Single Page Application with dynamic DOM mutations.
- Built real-time password strength analyzer with visual progress feedback.
- Verified with automated tests (test_task3.js and test_task4.js passed with code 0).